Information Management
Assignment 2

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What are the differences between relation schema, relation and instance? Give an example using the university database to illustrate. (Part I)

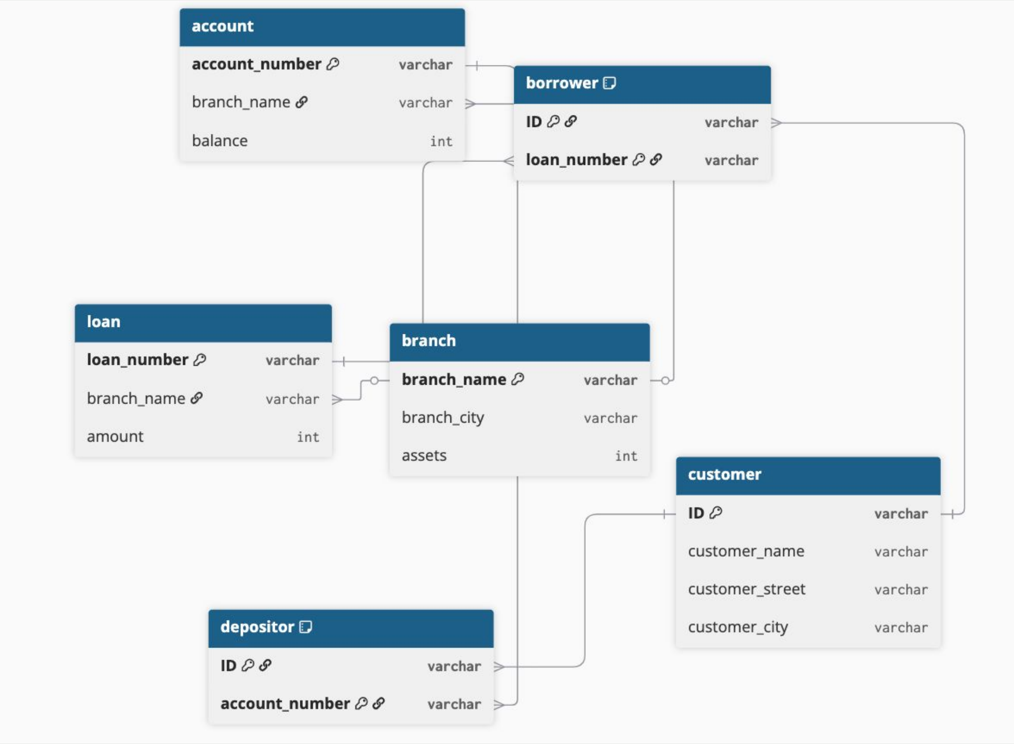
- A relational schema in my understanding is the the whole base logic and structure of the database you create and based on that you design your whole database. I think of it like the way an architect needs a blueprint of a building before you actually build a building or even start the base of the building. I like to think more in the context of political science research where its always said that good research anchors to a strong theoretical framework and its not about fancy methods as much as the theory.
- A relational schema is static and includes attributes and what type of data comprises within each attribute.
- Based on the uni example: Instructor (includes attributes such as ID, Name, Department name, courses, salary, etc)

What are the differences between relation schema, relation and instance? Give an example using the university database to illustrate. (Part II)

- Unlike the schema which tells you how multiple tables are structurally designed and connected a relation is just simply a format table structure that consists of a set of tuples. These rows never repeat themselves.
- Continuing the same example the relation of instructor will consist of multiple tuples with names of professors.
- Then instance is like a screenshot of the data in the relation of the schema at angy given point of time. This changes because we can at any time add, deplete or modify data within these relations. For example. How the utd coursebook looks after every semester changes because they add new courses and instructors and / or classes.

Q2 Bank schema

- Made on dbdiagram.io



Q3 Bank schema

Table Definitions with Primary Keys

- branch (branch_name, branch_city, assets)
- customer (ID, customer_name, customer_street, customer_city)
- loan (loan_number, branch_name, amount)
- borrower (ID, loan_number) *Composite key used because loans can have multiple customers*
- account (account_number, branch_name, balance)
- depositor (ID, account_number) *Composite key used because accounts can have multiple customers*

Identified Foreign Keys

- loan(branch_name) references branch(branch_name)
- account(branch_name) references branch(branch_name)
- borrower(ID) references customer(ID)
- borrower(loan_number) references loan(loan_number)
- depositor(ID) references customer(ID)
- depositor(account_number) references account(account_number)
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Q4. Describe two ways artificial intelligence or LLM can assist in managing or querying a database. In your answer, briefly explain how each method improves efficiency or accuracy compared to traditional (non-AI) approaches. (3–5 sentences

- LLMs allow non-technical users to query databases using plain English, which improves efficiency by removing the syntax barrier of traditional SQL.
- AI can automatically index or do control F
- Better retrieval by searching through AI